Mathematical Analysis 2

Seminar 11

Line integrals (2)

Integrals w.r.t. coordinates / over vector fields

$\vec{F} = (P, Q, R)$; $\vec{F} = \vec{F}(x, y, z)$

$(C) : \vec{r} = \vec{r}(t) = x(t)\vec{i} + y(t)\vec{j} + z(t)\vec{k}$, $t \in [a, b]$

$$\int_C \vec{F} \bullet d\vec{r} = \int_C P\,dx + Q\,dy + R\,dz = \int_C \underbrace{\left(\vec{F} \bullet \frac{d\vec{r}}{ds}\right)}_{\substack{\text{scalar value} \\ \text{(scalar product)}}} ds$$

$$= \int_a^b \Big(P\big(x(t), y(t), z(t)\big)\, x'(t) + Q(\dots)\, y'(t) + R(\dots)\, z'(t) \Big)\, dt$$

WinPlot animation in a separate file

$\boxed{\textbf{Ex. 1}}$ Compute $\displaystyle\int_C \overrightarrow{F}\cdot d\overrightarrow{r}$, where $\overrightarrow{F} = \underbrace{e^{2x}}_{P}\,\overrightarrow{i} + \underbrace{z(y+1)}_{Q}\,\overrightarrow{j} + \underbrace{z^3}_{R}\,\overrightarrow{k}$ and

$(C): \overrightarrow{r}(t) = \underbrace{t^3}_{x}\,\overrightarrow{i} + \underbrace{(1-3t)}_{y}\,\overrightarrow{j} + \underbrace{e^t}_{z}\,\overrightarrow{k}$, $t \in [0,1]$.

$$I = \int_C P\,dx + Q\,dy + R\,dz = \int_C e^{2x}dx + z(y+1)dy + z^3 dz = \int_0^1 \left(e^{2t^3}\cdot 3t^2 + e^t(2-3t)(-3) + e^{3t}e^t\right)dt$$

$$= \int_0^1 \left(3t^2\cdot e^{2\boxed{t^3}} + e^{4t} + e^t(-6+9t)\right)dt = \frac{1}{2}e^{2t^3}\Big|_0^1 + \frac{1}{4}e^{4t}\Big|_0^1 + e^t(9t-6)\Big|_0^1 - \int_0^1 e^t\cdot 9\,dt =$$

$$= \frac{1}{2}(e^2-1) + \frac{1}{4}(e^4-1) + 3e - (-6) - 9e^t\Big|_0^1 = \frac{1}{2}(e^2-1) + \frac{1}{4}(e^4-1) + 3e + 6 - 9e + 9 = \ldots$$

Ex. 2 How much work is done by the vector field $\overrightarrow{F} = x\,\overrightarrow{i} - y\,\overrightarrow{j} + z\,\overrightarrow{k}$ along the line segment $[AB]$, from $A(2,3,1)$ to $B(-1,2,1)$?

$$\int_C \vec{F}\cdot d\vec{r} = \int_{[AB]} x\,dx - y\,dy + z\,dz.$$

$[AB]$: $\vec{r}(t) = (1-t)\,\vec{r}_A + t\,\vec{r}_B$, $t\in[0,1]$; $t=0 \Rightarrow \vec{r}(0)=\vec{r}_A$; $t=1 \Rightarrow \vec{r}(1)=\vec{r}_B$

$$\begin{cases} x(t) = (1-t)\cdot 2 + t\cdot(-1) = -3t+2 \\ y(t) = (1-t)\cdot 3 + t\cdot 2 = 3-t \\ z(t) = (1-t)1 + t\cdot 1 = 1 \end{cases} \quad \begin{array}{|l} x' = -3 \\ y' = -1 \\ z' = 0 \end{array}$$

$$I = \int_0^1 \Big((-3t+2)(-3) - (3-t)(-1) + 0\Big)\,dt = \int_0^1 (9t-6+3-t)\,dt = \int_0^1 (8t-3)\,dt = (4t^2-3t)\Big|_0^1 = 1$$

2^{nd} method: $x\,dx - y\,dy + z\,dz = d\left(\frac{x^2}{2}\right) - d\left(\frac{y^2}{2}\right) + d\left(\frac{z^2}{2}\right) = d\left(\frac{x^2-y^2+z^2}{2}\right)$

$\vec{F} = \text{grad}\Big(\underbrace{\frac{x^2-y^2+z^2}{2}}_{\varphi}\Big)$ (gradient field) (conservative field) $\left(\vec{F} = (\varphi'_x, \varphi'_y, \varphi'_z)\right)$

$$I = \varphi(B) - \varphi(A) = \varphi(-1,2,1) - \varphi(2,3,1) = \frac{1-4+1}{2} - \frac{4-9+1}{2} = -1-(-2) = 1.$$

Ex. 3 Show that $\vec{F} = \underbrace{(y^2\cos x + z^3)}_{P}\vec{i} \underbrace{-(4-2y\sin x)}_{Q}\vec{j} + \underbrace{3xz^2}_{R}\vec{k}$ is a gradient field and find a primitive ϕ of $\vec{F}$ ($\operatorname{grad}\phi = \vec{F}$).

$\operatorname{curl}\vec{F} = \vec{0} \iff \boxed{\begin{matrix} P'_y = Q'_x \\ 2y\cdot\cos x \ \checkmark \end{matrix}}\ ;\ \boxed{\begin{matrix} P'_z = R'_x \\ 3z^2 \ \checkmark \end{matrix}}\ ;\ \boxed{\begin{matrix} Q'_z = R'_y \\ 0 \ \checkmark \end{matrix}}$

Finding φ:

$$P\,dx + Q\,dy + R\,dz = (y^2\cdot\cos x + z^3)\,dx + (-4+2y\cdot\sin x)\,dy + 3xz^2\,dz =$$

$$= y^2\cdot\cos x\,dx + z^3\,dx - 4\,dy + \sin x\cdot 2y\,dy + x\cdot 3z^2\,dz =$$

$$= y^2\cdot d(\sin x) + z^3\,dx + d(-4y) + \sin x\cdot d(y^2) + x\cdot d(z^3) =$$

$$= d(y^2\cdot\sin x) + d(xz^3) + d(-4y)$$

$$= d(y^2\cdot\sin x + xz^3 - 4y) = d\varphi$$

$\varphi(x,y,z) = y^2\cdot\sin x + xz^3 - 4y + C$, $C\in\mathbb{R}$

$\boxed{\text{Or}}$

$\varphi'_x = P$; $\varphi'_y = Q$; $\varphi'_z = R$; $\varphi'_x = y^2\cos x + z^3 \ \Big|\int dx \Rightarrow \varphi = y^2\sin x + xz^3 + C(y,z)\ \Big|\frac{\partial}{\partial y} \Rightarrow$

$\Rightarrow \left.\begin{matrix}\varphi'_y = 2y\sin x + 0 + C'_y \\ = Q = 2y\cdot\sin x - 4\end{matrix}\right\} \Rightarrow C'_y = -4 \ \Big|\int dy \Rightarrow C(y,z) = -4y + C(z) \Rightarrow$

$\Rightarrow \varphi = y^2\sin x + xz^3 - 4y + C(z)\ \Big|\frac{\partial}{\partial z} \Rightarrow \left.\begin{matrix}\varphi'_z = 3xz^2 + C'(z) \\ = R = 3xz^2\end{matrix}\right\} \Rightarrow C'(z) = 0 \ \Big|\int dz \Rightarrow C(z) = C \in\mathbb{R}$

$\varphi = \ldots$

$\boxed{\textbf{Ex. 4}}$ Compute $\displaystyle\int_C \overbrace{x}^{Q}\,\mathrm{d}y + \overbrace{y}^{P}\,\mathrm{d}x + \overbrace{\frac{z}{z^2+1}}^{R}\,\mathrm{d}z$,

$(C): \vec{r}(t) = \underbrace{2a\cos^2 t}_{x(t)}\,\vec{i} + \underbrace{a\sin 2t}_{y(t)}\,\vec{j} + \underbrace{t}_{z(t)}\,\vec{k}$, $\quad t \in \left[0, \frac{\pi}{2}\right]$, $\quad a > 0$ fixed.

? $P'_y = Q'_x = 1$; $P'_z = R'_x = 0$; $Q'_z = R'_y = 0$ ✓

curl $\vec{F} = \vec{0}$

$\Rightarrow \vec{F}$ is a gradient field.

Directly, we can find φ :

$\underbrace{x\,dy + y\,dx}$ $+ \frac{z}{z^2+1}\,dz = d(xy) + d\left(\frac{1}{2}\ln(z^2+1)\right) = d\Big(\underbrace{xy + \tfrac{1}{2}\ln(z^2+1)}_{\varphi}\Big)$

$I = \varphi\left(\vec{r}\left(\frac{\pi}{2}\right)\right) - \varphi\left(\vec{r}(0)\right) = \varphi\left(0, 0, \frac{\pi}{2}\right) - \varphi(2a, 0, 0) =$

$= \boxed{\frac{1}{2}\ln\left(\frac{\pi^2}{4} + 1\right)}$